C 20661 **(Pages : 2)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS—UG)

Physics/Applied Physics

PHY 6B 13/APH 6B 13—RELATIVISTIC MECHANICS AND ASTROPHYSICS

(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A

Answer at least **eight** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 24.

1. What is meant by transformation equations in relativity ?
2. Why did Einstein assume that the speed of light in vacuum is a universal constant ?
3. State the Cosmological principle.
4. Is the white dwarf a stable equilibrium state ? Explain why.
5. Give the postulates of special relativity.
6. Define luminosity of a star. Is the luminosity of a star related to its temperature ?
7. Sirius has a parallax of 0.379 arcseconds. How much parsec is its distance from the Earth ?
8. What is CMBR ?
9. Explain Hubble's law for the expansion of the universe.
10. Explore the features of AGN. Why are they called so ?
11. What are Binary stars ?
12. Mention the features of Population II stars ?

(8 × 3 = 24 marks)

Turn over

Section B (Paragraph/Problem Type)

Answer at least **five** *questions.*
Each question carries 5 *marks.*
All questions can be attended.
Overall Ceiling 25.

13. Show that in the non-relativistic limit, Lorentz transformations reduces to Galilean transformations.

14. (a) If a star's brightness is increased by 45,000 times, how much is the change in its magnitude ?

    (b) Faint Brown dwarf stars have absolute magnitudes of around 17.5. How much fainter they are compared to Sun ?

15. Mention the important features of Cepheid variables. Why are they used as distance indicators ?

16. Discuss the standard big-bang theory for the origin of the universe.

17. A particle is traveling through the Earth's atmosphere at a speed of 0.6c. To an Earth-bound observer, the distance it travels is 1.50 km. How far does the particle travel in the particle's frame of reference ?

18. Discuss the classification of galaxies.

19. What kinds of objects make up the dark matter ?

(5 × 5 = 25 marks)

Section C (Essay Type)

Answer any **one** *question.*
The question carries 11 *marks.*

20. Derive the law of addition of velocities using Lorentz transformation equations.

21. Explain the star formation and life cycle of stars.

(1 × 11 = 11 marks)